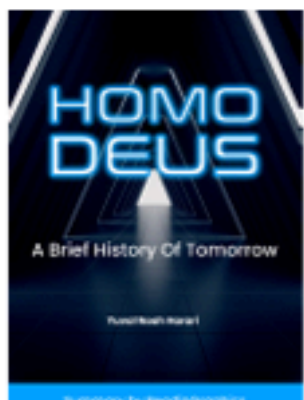




HOMO DEUS

A Brief History of Tomorrow

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The Big “So What”



Today, Homo sapiens have dominated the world, and brought famine, plague and war under control. Where will we channel our ambitions next, and what could happen as we progressively tweak our genes and incorporate artificial intelligence into our world? **In this book, Harari examines our past, present and future, and highlights some difficult questions and choices that could shape our collective future.**

Introduction



Over the last century, humans (*Homo sapiens*) have conquered our biggest enemies—famine, plague and war—and ruled the planet as the dominant species.

- In ancient Egypt or medieval India, severe droughts could wipe out 5-10% of the population. The 2 world wars killed about 100 million people, while the “Spanish Flu” wiped out about a third of the global population.
- Today, while millions of people still suffer from hunger or malnutrition, few actually starve to death. Diseases like SARS, Ebola and AIDS are contained, war is rare and people value political, social and economic stability. In fact, more people die today from obesity than from starvation; from old age than from infectious diseases; and from suicide than warfare.

In a healthy, prosperous and harmonious world, what will humans focus our attention on? Will our desire for growth, eternal youth and happiness ironically bring an end to our species? Harari presents his thoughts and arguments through a range of questions, examples, facts and theories, and highlights several

KEY QUOTES

“Given our past record and our current values, we are likely to reach out for bliss, divinity and immortality—even if it kills us.”

“Famine, plague and war... are no longer unavoidable tragedies...Instead, they have become manageable challenges.”

“Previous generations thought about peace as the temporary absence of war. Today we think about peace as the implausibility of war.”

“If incidences of famine, plague and war are decreasing, something is bound to take their place on the human agenda.”

questions that could shape our collective outcomes—before it's too late. In this summary, we've distilled and organized the key ideas into 2 parts: "How humans conquered the world" and "Where are humans headed?".

Part1: How Humans Conquered the World



Homo Sapiens: the Dominant Species



Homo sapiens are the first species to single-handedly change the world's ecology in 4 billion years since life first appeared on Earth. The appearance of the first Homo sapiens quickly drove other Homo species into extinction. Today, 90% of the large animals around the world are either humans or our domesticated animals. There are only 200,000 wild wolves vs 400 million domesticated dogs, 900,000 African buffaloes vs 1.5 billion domesticated cows.

Over time, our beliefs, ideologies, and relationship with other living creatures have evolved:



Archaic Hunter-gatherers ↔ Animism. Our first sapiens ancestors were probably animists, i.e. they saw humans, animals, plants, rivers and even stones, as an integral part of the ecosystem and landscape.



Agricultural Revolution ↔ Theism. Theists argue that Gods created the universe, favoured humans over other species, and provided us with all we needed. During the Agricultural Revolution, Theism was popularized, and mankind started to see ourselves as a special species. Animals were domesticated, and offered as sacrifices in return for the Gods' blessings. In ancient Egypt, medieval China and biblical Israel, certain "classes" of humans were also treated like "lesser animals", and were enslaved and tortured.



Scientific Revolution ↔ Humanism. As mankind developed new understanding of chemistry, biology and physics, Gods were slowly removed from the picture. Humans can now heal ourselves, improve crops, forecast the weather—all without the Gods' assistance. With science and antibiotics, we could breed thousands of cows, chickens, pigs etc. and pack them into cramped spaces without

KEY QUOTES

"Instead of fearing asteroids, we should fear ourselves. For Homo sapiens has rewritten the rules of the game."

"The farm thus became the prototype of new societies, complete with puffed-up masters, inferior races fit for exploitation, wild beasts ripe for extermination, and a great God above that gives His blessing to the entire arrangement."

"We can be quite certain that humans will make a bid for divinity, because humans have many reasons to desire such an upgrade, and many ways to achieve it."

major epidemics. In the eyes of other animals, humans may well be gods.



Homo Deus ↔ Techno-ism? With plague, famine and war under control, Harari believes that humans will seek to attain eternal youth and bliss. To upgrade ourselves, we'll tweak our biology and develop artificial intelligence (AI). Yet, are these modified super-humans, Homo Deus, really human? We'll discuss this later in Part 2.

Homo Sapiens: Who are We, Really?



Humans want to believe that we're special, and the most commonly used reasons are: humans have eternal souls and/or consciousness, while animals have neither.



Monotheists argue that humans have eternal souls; when we die, our souls move on to salvation or damnation. However, there has been no way to prove that humans indeed have a soul.



Another argument is that Homo Sapiens have a conscious mind. We can perceive subjective experiences (including sensation and desire) and combine them to form a stream of consciousness, to feel "overwhelmed", "fulfilled", etc.

- Consciousness is proven as a real phenomenon. When something happens, the body produces electrons and biochemical responses, to give us sensations and desires. Thus, by examining the brain, scientists can tell if someone is awake or sleeping, and if he's afraid or angry; we can even differentiate between the brain neurons associated with different ideas.
- However, as of 2016 when the book was published, we can't explain how consciousness is formed, nor what role it plays in our evolution. Thus, we can't assume that animals and inorganic beings (e.g. computers) can't be or become conscious.

Animals like monkeys and dogs certainly demonstrate intelligence and awareness. Lab rats were distressed by other rats being held captive and found ways to free them. A chimpanzee at a Swedish zoo could devise various strategies to discreetly hide objects and throw them at

KEY QUOTES

"The belief that humans have eternal souls... is a central pillar of our legal, political and economic system."

"From an evolutionary perspective, the closest thing we have to a human essence is our DNA... This terrifies large numbers of people, who prefer to reject the theory of evolution rather than give up their souls."

"Desires are nothing but a pattern of firing neurons."

"Scientists don't know how a collection of electric brain signals creates subjective experiences."

visitors. In 2015, both New Zealand and Canada officially recognized animals as sentient beings, and more countries may follow suit.

Can inorganic beings like computers and robots get sophisticated enough to develop consciousness? The Turing Test says that, if you were to communicate simultaneously with a computer and human, and are unable to distinguish which is which, then the computer has a mind. At the end, it's what we *think* is real that matters. If a computer *seems* to think and act like a human, we're likely to believe it has consciousness.

Harari argues that all organisms are algorithms, including humans, i.e. we use a methodical set of steps to make calculations, solve problems and reach decisions. Our brains make rapid decisions daily, using complex algorithms that combine an analysis of our surroundings, past experiences, motives etc. That's how you assess if you can dash across the road before an oncoming car reaches you, how a rabbit decides if it's better to dash across a field or try to hide from a nearby fox.



Many of these algorithms are woven into our genetic makeup from millions of years of evolution. For example, mammals are built to care for their offspring, and their young intuitively suckle and bond with the adults. Human children who grow up in loving, secure families are more likely to become confident and secure adults. Baby monkeys separated from their mothers became attached to soft, fluffy objects like dolls/blankets, and those without affection grew up to be dysfunctional adults.



Theism gave humans justifications for domesticating other animals. However, it doesn't change the fact that animals have social, emotional and primal instincts, just like us.

- For example, wild boars are intelligent, social mammals that need to play and mingle as a group. Their descendants, domesticated pigs, are also smart with emotional/social needs. Yet, sows in industrial farms often live their entire lives being locked in tiny crates; each time they give birth, they're separated from their piglets in just 2-4 weeks (a fraction of the natural 10-20 weeks' nursing cycle).
- Humans may be more intelligent and powerful than other animals, but does that justify how we treat them? If a

KEY QUOTES

"It won't matter whether computers will actually be conscious or not. It will matter only what people think about it."

"The Agricultural Revolution gave humans the power to ensure the survival and reproduction of domesticated animals while ignoring their subjective needs."

"What makes the fate of domesticated farm animals particularly harsh is not just the way they die, but above all the way they live."

superior species Homo Deus were to emerge, will they treat Homo Sapiens the same way?

If humans are algorithms, don't have eternal souls nor special consciousness, then how did we conquer the world? Harari argues that we became the dominant species because of our unique ability to organize ourselves flexibly and in large numbers.

Homo Sapiens: Giving Meaning to the World



Our ability to imagine, create stories and meaning allows us to organize ourselves unlike other species.

- Bees and ants work well in large numbers, but they can't change the structure of their colonies easily. Other animals like elephants and chimpanzees can cooperate flexibly, but only in small groups. Humans have the unique ability to cooperate flexibly and in large numbers, because we're the only species that can create meaning, invent and spread stories.
- Human communities can create norms and shared beliefs, e.g. "if you sacrifice 10 goats, you'll get a good harvest", or "if you kill someone, you'll go to hell". Since people treat the beliefs as reality, we can predict how strangers will behave, and organize people on a large scale, to create legal systems, stock-markets, global trade, etc.



Same algorithm, different stories. Human stories change, but the algorithms behind how societies are organized remain the same.

- Our social algorithm goes like this: Decisions are made by a network of people, fronted by a figurehead, and carried out in the name of a bigger idea that gives meaning to the masses.
- However, the story behind the algorithm changes, e.g. ancient Egyptians believed in Pharaohs and Gods, while modern people believe in brands like Google. Yet, 12,000 years ago, Gods functioned like big corporations—they owned land, paid salaries and built infrastructure. People prayed to the Pharaoh to alleviate floods, but the real solutions came from coordinated efforts by temple priests

KEY QUOTES

"To the best of our knowledge, only Sapiens can cooperate in very flexible ways with countless numbers of strangers."

"All large-scale human cooperation is ultimately based on our belief in imagined orders."

"If we want to understand our future...we must also decipher the fictions that give meaning to the world."

"No other animal can stand up to us, not because they lack a soul or mind, but because they lack the necessary imagination."

"Fictions enable us to cooperate better. The price we pay is that these same fictions also determine the goals of our cooperation."

and the masses, to build dams and canals. In many ways, the Pharaoh was no more than a powerful brand.

- What matters to people in one era can become useless in another. Zeus and Hera were all-powerful in the past, but just myths today. Likewise, our modern-day ideas of human rights and democracy may seem strange to people a few centuries later.



Written language is powerful; it seems more real, can spread knowledge, shape reality and misrepresent truths. In fact, when text and reality clashes, reality sometimes loses.

- For example, in the late 19th century, European powers divided up the African territories, but the borders drawn on paper didn't reflect the realities in Africa. When the discrepancies were discovered years later, the authorities decided to stick to the paper agreement, which became the new reality.
- Computer algorithms will make it even harder to differentiate fiction from reality, and amplify the impact of stories at an increasing pace.



Human societies are held together by a web of meaning.

- There are 2 types of reality. An objective reality affects everyone, whether they believe it or not, e.g. the law of gravity. A subjective reality depends on personal beliefs and feelings, e.g. our beliefs about God or money.
- Our shared stories are reinforced in a self-perpetuating web, e.g. what it means get married in a church, or vote in an election. The more people believe them, the more they're practiced and the beliefs are reinforced.
- You can set up a system for any goal (e.g. equality, preservation of life, making money), and create stories and criteria to justify how well it's doing. For example, a school with higher average marks is considered a better school, even if marks may not reflect the quality of education.



Science vs Religion. Science and religion do not clash; instead, they complement each other.

KEY QUOTES

"History isn't a single narrative, but thousands of alternative narratives. Whenever we choose to tell one, we are also choosing to silence others."

"Written language may have been conceived as a modest way of describing reality, but it gradually became a powerful way to reshape reality."

"Stories are just tools. They should not become our goals or our yardsticks."

"We want to believe that our lives have some objective meaning... Yet in truth the lives of most people have meaning only within the network of stories they tell one another."

"Instead of changing their stories to fit reality, (bureaucracies) can change reality to fit their stories."

- Science gives us *power*, using fact-based solutions to produce food, cure diseases, and fight wars. However, it can't make ethical judgments to tell us *how* to act.
- Religions are basically rules that're attributed to a wider moral law, and may not necessarily relate to Gods. They provide *order*, by offering ethical values, subjective goals, and guidelines on what we should do. Most *religious stories* have 3 parts: (a) ethical judgements e.g. "human life is sacred", (b) factual statements e.g. "human life starts at the point of conception", and (c) a merger of the 2 parts to form practical guidelines e.g. "you must not allow abortion once you've conceived".

While science can't make ethical judgements, it can *validate* if religious facts are accurate. For example, many Christians and Jews believe that the book of Leviticus was dictated by God to Moses, and no word was added/deleted since. However, scientific studies suggest that the Bible is a collection of texts by different human authors, written years after the events described, and compiled into a holy bible long after biblical times.

- Science and religion can and should co-exist. **Harari argues that to stay relevant, religions must understand modern technology, so they can guide people on real-world issues,** like: What can we do when artificial intelligence replaces our jobs? Where do we draw the line for gene enhancements and life extensions? Traditional religions that fail to do so may well fade into oblivion.

With the power to create meaning and scientific solutions, humans can choose to do anything. The purpose we serve, the stories we choose, and our ability to separate fact from fiction, will shape our collective social, political and economic outcomes.

Part 2: Where are Humans Headed as a Species?



In the book, Harari touches on various developments and perspectives that are shaping our future, and the dilemmas that they bring.

KEY QUOTES

"Religion is created by humans rather than gods, and it is defined by its social function rather than by the existence of deities."

"Religion is any all-encompassing story that confers superhuman legitimacy on human laws, norms and values."

Capitalism and Growth



Capitalism has brought mankind amazing growth, progress, power and control. However, **we're now faced with a dilemma: continuing our growth will likely lead to a total ecological collapse, but stopping the wheel of growth can crash our global systems.**



Credit is a key building block of our modern economy. It's given in faith that someone will deliver better results in future, and return the credit with interest. Farmers take loans to buy new seeds and fields, people invest their money in stocks, etc. As new ventures succeed, people have even more faith in the future, credits become more accessible and our economy grows.



Capitalism assumes the wheel of growth will spin constantly and indefinitely. Yet, can the ecology support our rapid growth? Although science has constantly delivered new sources of energy, raw materials, and improved production methods, we face a real threat of total **ecological collapse**, which could wipe out all our systems, and even mankind itself.

- We're stuck in a **catch-22 situation**: we want faster progress and growth, yet we must slow down to avoid an ecological collapse.
- Since the rich want more money, and the poor are the first to suffer in an economic slump, **growth continues to trump ecology**. Beijing's air pollution now supports a growing air-purification market; the rich buy air-purifiers and build protective structures over their buildings, while air quality worsens. For all our talk of global warming, greenhouse gas emissions continued to grow at 2.2% annually from 2000-2010. The USA (the top global polluter) refused to ratify the 1997 Kyoto protocol for emissions reduction.

The Humanist Revolution



Humanism is the dominant religion today—it places humans at the centre of the universe, and judges if something is good or bad based on the impact on us. We no longer depend on the Gods and religious scriptures to tell us what's good or right. Instead, we believe in our free will, and depend on our inner feelings and desires to guide our

KEY QUOTES

"Credit is the economic manifestation of trust."

"An ecological meltdown will cause economic ruin, political turmoil, a fall in human standards of living, and might threaten the very existence of human civilisation."

"We talk a lot about global warming, but in practice humankind is unwilling to make the serious economic, social or political sacrifices necessary to stop this catastrophe."

decisions. Even those who believe in God choose to do so, because their inner voice tells them God exists.

 There are 3 main sub-branches to Humanism.

- Socialist Humanism advocates having strong collective institutions (e.g. unions) to consider the overall impact on our societies/world.
- Evolutionary Humanism advocates leaving things to natural selection—let the superior Homo sapiens win, so mankind gets stronger as a species.
- Liberalism advocates that every human being should be free to express his/her unique inner voice. As of 2016, Liberalism is in the lead, manifested in the ideologies of human rights, individualism, democracy and the free market.

 However, **Liberalism is built on 2 key ideas, both of which are myths:**

- It assumes humans have free will and thus should have free choice. But in reality, we can't control our thoughts/feelings. Our brains' electro-chemical impulses can also be manipulated, and such interventions are already being used in medical treatments and performance enhancements.
- It assumes we are holistic beings with a single mind. The truth is, we are full of inner conflicts and can literally be of many minds. Kahneman found that we have 2 selves: an *experiencing self* (which feels our full experiences) and a *remembering self* (which recalls only parts of our experiences, mixes them with other memories or impressions, and develops stories about them). Our decisions are made by the remembering self.

 **Harari believes Liberalism will die, and humans will lose our value as unique individuals, with 3 possible developments:**

- As computers become more intelligent, humans lose our value altogether. Google's self-driving cars are already a reality. Robots, 3D printers and automations have replaced manual workers, and are slowly replacing service jobs. IBMS' AI system, Watson, can even diagnose diseases faster and more effectively than humans. While AI algorithms are

KEY QUOTES

"In 2016 the world is dominated by the liberal package of individualism, human rights, democracy and the free market."

"If organisms indeed lack free will, it implies that we can manipulate and even control their desires using drugs, genetic engineering or direct brain stimulation."

"If I had more than one authentic voice, how would I know which voice to heed?"

"When mindless algorithms are able to teach, diagnose and design better than humans, what will we do?"

still flawed, these are technical issues which can be fixed. Since organisms are algorithms, and it doesn't matter what material the algorithmic calculator is made of, it's possible for non-organic algorithms to someday replicate or surpass humans.

- We no longer assert our individualism but stay relevant as a collective species. We're already depending more and more on external algorithms (e.g. wearable technology, mobile apps) to make our decisions, including travel routes, purchases, customizing our diet plans or medication dosage. The more data/access we give to these algorithms, the more accurate their predictions, until one day they can know us better than we know ourselves. Once that happens, we're likely to leave our decisions to the algorithms, rather than look inward to our "true selves". The algorithms may even manipulate how we feel and think, while retaining our illusion of free will.
- A small group of elite, upgraded superhumans may emerge to dominate other humans using technology and computer algorithms. The overall level of healthcare, productivity etc. will probably become better, but the gap between the elites and masses will be significantly widened.

The New Human Agenda



Besides our need for never-ending growth, humans also have ambitions for immortality (eternal youth) and lasting happiness.

∞ Eternal youth. In the past, people accept death as part of our spiritual metaphysical experience. Today, life is increasingly seen as a fundamental human right. We seek *medical solutions* for everything from cancer to Alzheimer's, and find ways to *reduce deaths* from crime, accidents etc. Even Google Ventures is investing in life sciences and *life-extension*.

🌿 Happiness is now also seen as a basic human right, and people demand the freedom to pursue their bliss. Yet, finding true happiness isn't easy. On a *psychological level*, rising expectations make it hard for people to feel content. On a *biological level*, we can induce positive feelings by manipulating our biochemical system (using drugs, alcohol, mental stimulants, etc.). However, these effects don't last, and

KEY QUOTES

"The same technologies that can upgrade humans into gods might also make humans irrelevant."

"Humans will no longer be autonomous entities... Reality will be a mesh of biochemical and electronic algorithms, without clear borders, and without individual hubs."

"As long as people die of something, we will strive to overcome it."

"If science is right and our happiness is determined by our biochemistry system, then the only way to ensure lasting contentment is by rigging this system."

we're likely to seek new ways to tweak our biology to make the pleasurable effects more permanent.

A New World of Homo Deus and Techno-Religions



As Liberalism gets ousted, and humans push for ever more growth and biological upgrades, new techno-religions could emerge:



Techno-humanism still recognizes humanity at the apex of creation, but uses technology to create the superior Homo Deus with eternal youth and happiness. This may be done through:

- Biological engineering, e.g. rewriting our genetic codes, rewiring our brains, or growing new body parts.
- Cyborg engineering, to merge our organic bodies with non-organic devices, e.g. bionic hands or artificial eyes.
- Creation of non-organic beings, with minds that're so different, we can't fathom how they'll work, just like how our cavemen ancestors couldn't imagine stock markets or smart phones.

In the process of upgrading our bodies and brains, we may end up with a new species of superhumans with an entirely different mind.



Data religion: Dataism is about using computer algorithms to turn massive amounts of information into knowledge and wisdom.

- By tracking what people search online, Google can already detect new epidemics faster than health organizations. Imagine if we could capture the algorithms of organisms and entire ecologies (e.g. beehives, human cities, and forests) and input all the data into a central-processing system. What new outcomes, systems and structures will this give birth to?
- Homo sapiens may be better than other animals in absorbing and using algorithms, but we're likely to be obsolete in the world of Dataism. Yet, the more data is connected, the more

KEY QUOTES

"In pursuit of health, happiness and power, humans will gradually change first one of their features and then another, and another, until they will no longer be human."

"Technological progress has a very different agenda. It doesn't want to listen to our inner voices. It wants to control them."

"As the global dataprocessing system becomes all-knowing, and all-powerful, so connecting to the system becomes the source of all meaning."

we can't resist plugging into the system, further accelerating the trend.

Where do we go from here?



Most of Harari's ideas above are not prophecies, but a projection of our possible futures, based on lessons from our own history. Our choices today are what will define our future outcomes.



The paradox of knowledge. In the 19th century, Karl Marx predicted the collapse of capitalism. His diagnosis sparked awareness and changes over the years, contributing to an opposite outcome to his prediction. By sharing the possible futures of mankind, Harari hopes to shape our knowledge, understanding, choices and actions, so we can collectively change the future.



Nonetheless, reversing these trends will not be easy, because:

- Everything is interconnected, and there's no single brake to pull.
- There's no clear line between "good" vs "bad", healing vs upgrading. For example, Viagra started out as a treatment for blood pressure, and plastic surgery emerged after World War I to treat facial injuries.
- We face real dilemmas e.g. the need for growth vs ecological preservation, our desire for technological advancements vs the concern of losing control.
- In our quest for power/control, we've given up the wider meaning/ purpose that held societies together for millennia. What should be the new higher purpose that guides our collective choices today, and can we rally ourselves around a common goal?

There are no easy answers to these issues. Harari hopes that this book will spark discussion and debate to make us more conscious of the technologies we use and the choices we make, before the future becomes inevitable.

KEY QUOTES

"This prediction is less of a prophecy and more a way of discussing our present choices."

"Since no one understands the system any more, no one can stop it."

"Healing is the initial justification for every upgrade."

"To attain real happiness, humans need to slow down the pursuit of pleasant sensations, not accelerate it."